

High-Efficiency Encapsulation of Pt Nanoparticles into the Channel of Carbon Nanotubes as an Enhanced Electrocatalyst for Methanol Oxidation

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Abstract: Pt-based nanostructures serving as anode catalysts for the methanol oxidation reaction (MOR) have been widely studied for many years. Nevertheless, challenging issues such as poor reaction kinetics and the short-term stability of the MOR are the main drawbacks of such catalysts and limit their applications. Herein, we have developed a facile approach to encapsulate Pt nanoparticles (NPs) inside the

nanochannels of porous carbon nanotubes (CNTs; Pt-in-CNTs) as a new enhanced electrocatalytic material. The as-prepared CNTs offer simultaneously ordered diffusion channels for ions and a confinement effect for the NPs,

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which both facilitate the promotion of catalytic kinetics and avoid the Ostwald ripening of Pt NPs, thus leading to high activity and durable cycle life as an anode catalyst for MOR. This work provides a new approach for enhancing the stability and activity by optimizing the structure of the catalyst, and the Pt-in-CNTs represent the most durable catalysts ever reported for MOR.

Introduction

Direct methanol fuel cells (DMFCs) are regarded as promising future power sources, especially for mobile and portable applications, owing to advantages such as high energy density, low operating temperature, and simple processing.^[1–3] These kind of DMFCs require effective electrocatalysts for catalyzing the methanol oxidation reaction (MOR) with low overpotentials at the anode electrode.^[4–6] Generally, Pt is a commonly used anode catalyst for catalyzing the MOR.^[7] However, their sluggish kinetics and poor long-term stability for the MOR greatly limit the commercialization of the devices of DMFCs. Carbon nanotubes (CNTs) are one-dimensional (1D) carbon nanomaterials that have large surface area, excellent conductivity, and good chemical stability. CNTs have been explored extensively as a unique support of Pt NPs for promoting the activity of MOR through a syn-

ergetic effect between carbon and metal NPs, as demonstrated in the different catalysts of CNTs/Pt,^[8–11] CNTs/ionic liquid/Pt NPs,^[12,13] amorphous CNTs/Pt^[14], and so on. In these CNTs/Pt NPs catalyst systems, Pt NPs were deposited directly onto the external surface of CNTs through in situ chemical deposition^[12,13] or a self-assembly strategy.^[10] Despite the fact that these state-of-the-art CNTs/Pt catalysts show great potential for enhancing the MOR activity, in the corrosive reaction conditions of DMFCs, CNT/Pt catalysts tend to have very limited durability due to the relatively weak interaction between CNTs and Pt NPs.^[15] The big challenge of MOR, the long-term stability, can still not be addressed well, which becomes the main obstacle in the commercialization of DMFCs.

Recent studies revealed that Pt-based NPs or other metals confined inside the channel of CNTs can exhibit enhanced catalytic activity towards liquid- and gas-phase reactions relative to NPs outside CNTs owing to the strong confinement effect of CNTs.^[16–23] For instance, Serp and co-workers found that Pt/Ru NPs driven into CNTs result in a high activity and selectivity towards the hydrogenation of cinnamaldehyde.^[17] Li and co-workers introduced cinchonidine-modified Pt NPs inside CNTs, showing higher enantioselective hydrogenation of α -ketoesters conversion compared with the NPs outside CNTs.^[18] Despite the great potential in catalytic applications of these NP-in-CNTs, they have not been explored as a catalyst for enhancing the electrooxidation reaction of small molecules due to the present limited synthetic control. In searching for more robust and practical MOR catalysts with much improved stability and activity, developing a new synthetic method for the high-efficiency confinement of Pt NPs into CNTs is highly desira-

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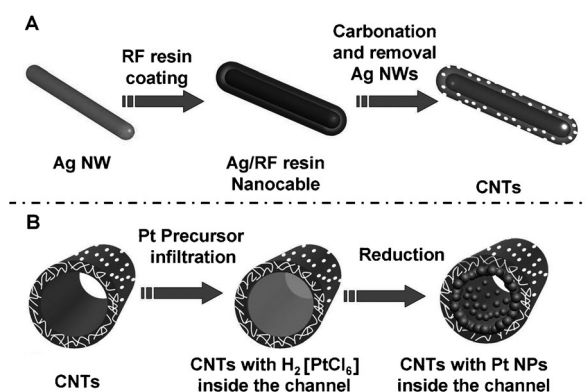
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ble, and such confining structure is expected to be very promising for enhancing the activity and stability of MOR.

Herein, we report a facile pathway for the high-efficiency encapsulation of Pt NPs inside the channel of CNTs (named Pt-in-CNTs) with high surface area and micropores by using a solution phase method. The as-obtained Pt-in-CNTs is a high-performance electrocatalyst for MOR in a solution of 0.5 M H_2SO_4 + 0.5 M methanol, with a much higher activity than those of Pt NPs supported on the surface of CNTs (named as Pt-out-CNTs) and the commercial Pt/C catalyst. The Pt-in-CNTs was also very stable for MOR and showed almost no activity change (9.1 % decrease) after 600 sequential potential cycles between -0.05 and 1 V versus Ag/AgCl (saturated KCl), whereas Pt-out-CNTs and commercial Pt/C showed a 44.6 and 73.5 % decrease in activity for MOR, respectively. The present work demonstrates the great potential of microporous CNTs as a nanocontainer to encapsulate NPs for enhancing the stability and activity for fuel cell reactions and other important catalytic reactions.

Results and Discussion

As is known, the as-prepared CNTs possess a 1D honeycomb-type carbon nanostructure. Annealing phenolic resins, derived from the polymerization of phenols (e.g., phenol, resorcinol) and aldehydes (e.g., formaldehyde, furfuraldehyde), at high temperature under N_2 protection, are commonly used for the production of carbon materials with many micropores in the carbon nanostructure.^[24,25] Such micropores will be advantageous for inward diffusion of the precursor into the channel of hollow nanocarbon such as CNTs, thus leading to NPs inside CNTs after the precursor was reduced. Scheme 1 shows the illustration procedure for the prepara-



Scheme 1. Schematic illustration procedure for the preparation of A) microporous CNTs and B) Pt-in-CNTs.

tion of microporous CNTs (A) and Pt NPs-in-CNTs (B). First, using Ag nanowires (NWs) as the template,^[26] well-defined Ag/resorcinol-formaldehyde (RF) resin coaxial nanocables were obtained by controlling the hydrolysis polymerization of resorcinol and formaldehyde onto the surface of

Ag NWs under ammonium catalysis. Then, by carbonizing Ag/RF resin nanocables at 850°C for 4 h under N_2 atmosphere and further eliminating the Ag core with 2 M HNO_3 , microporous CNTs were easily prepared (the details are described in the Experimental Section). As shown in Scheme 1B, the as-prepared microporous CNTs were then used as nanocontainers for encapsulating Pt NPs inside the channel of CNTs by an infiltration of the $[\text{PtCl}_6]^{2-}$ precursor into the inner cavity of CNTs based on the capillary forces of the micropores. After the reductant (ascorbic acid) was added into the above mixture, numerous small Pt NPs were spontaneously formed on the channel of CNTs, forming a NP network or nanodendrite-like structure, which is important for enhancing the MOR activity. It should be noted that for the effectively encapsulating Pt NPs inside CNTs, the microporous CNTs should be first functionalized with NH_2 group (CNTs- NH_2) through a typical sol-gel chemistry between oxygen-containing group of CNTs and (3-aminopropyl)triethoxysilane (APTES). Without this functionalization, Pt NPs could be only formed on the surface of CNTs.

Figure S1 A (the Supporting Information) shows the typical scanning electron microscopy (SEM) image of the as-prepared Ag NWs. They have a uniform diameter of approximately 50–80 nm and lengths up to tens of micrometers. After coating with RF, the diameter of coaxial nanocables increased to 170–200 nm with a dark Ag core and a light RF shell (Figure 1 A and Figure S1 B, the Supporting Information). The X-ray diffraction (XRD) pattern of Ag/RF coaxial nanocables displays four characteristic peaks of the face-centered-cubic phase of metallic silver with a calculated lattice constant $a = 4.1 \text{ \AA}$ (JCPDS Card File No. 04-0783; Figure S2 A, the Supporting Information). The thermogravimetric analysis (TGA) curve of Ag/RF shows a dis-

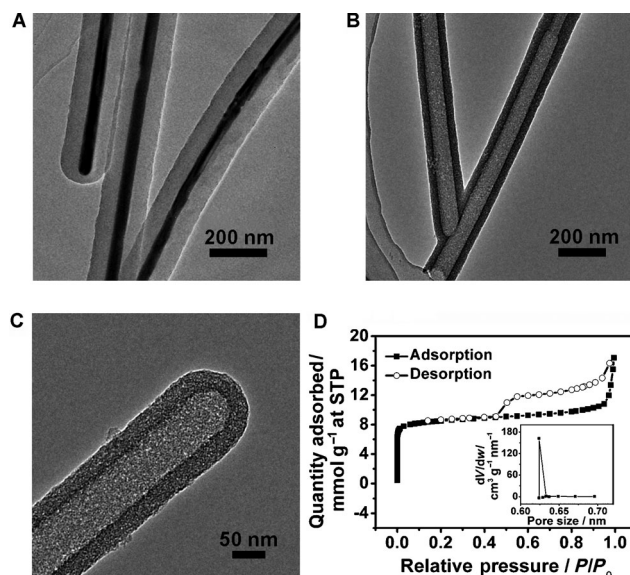


Figure 1. TEM images of A) Ag/RF resin nanocables and B, C) microporous CNTs. D) N_2 sorption isotherms and pore-size distribution (inset) of microporous CNTs in the microporous range. STP = standard temperature and pressure.

tinct weight loss starting around 350 °C, corresponding to the decomposition of RF resins (Figure S2B, the Supporting Information). The weight of Ag NWs in Ag/RF coaxial nanocables is calculated to be about 60 %. These results further demonstrate the existence of Ag NWs and organic polymer component in the Ag/RF resin nanocables.

Figure 1B and C, and Figure S1C (the Supporting Information) show the transmission electron microscopy (TEM) image of CNTs derived from removing the Ag core of the Ag/RF resin coaxial nanocables at different magnifications. Clearly, the well-defined CNTs with a closed end have an internal diameter of approximately 80 nm and a shell thickness of about 30 nm. The shell thickness of CNTs is smaller than that (about 60 nm) of the parent Ag/RF coaxial nanocables owing to the shrinkage during the carbonization process. The TGA curve of CNTs is shown in Figure S2C (the Supporting Information). No residue was found above 700 °C. Furthermore, inductively coupled plasma mass spectrometry (ICP-MS) data shows that no Ag exists in the CNTs. The results from both the TGA and ICP-MS again confirmed that no Ag remained in the as-prepared CNTs.

The magnified image in Figure 1C reveals that there are numerous worm-like micropores with the size of 0.5–1 nm in the wall of CNTs, which is a typical feature of RF resin-derived carbon materials.^[27,28] We further used the nitrogen sorption isotherms to determine the surface area and pore-size distribution, as shown in Figure 1D. The isotherm is of type I with a well-defined knee at low P/P_0 (0.0–0.1), implying the existence of micropores in CNTs, whereas the isotherm changes to type IV at $P/P_0 = 0.45$ –1, which is mainly attributed to the inner cylindrical channel of CNTs.^[29] The Brunauer–Emmett–Teller (BET) surface area and pore volume of the obtained CNTs are as high as 644 m² g^{−1} and 0.34 cm³ g^{−1}, respectively. The pore-size analysis for adsorption data, based on the Horvath–Kawazoe (H–K) method, clearly demonstrates that the prepared CNTs possessed micropores with an average diameter of 0.63 nm, being in accordance with the TEM results (Figure 1C and S1C, the Supporting Information). Such a particular structure is advantageous for offering a lot of entrances for the precursors to penetrate into the cavity of the CNTs. Raman spectroscopy of CNTs is shown in Figure S2C (the Supporting Information). We found that the ratio of I_D to I_G is 0.96, which is lower than that of graphene.^[30,31] This indicates that our prepared CNTs have a graphitic structure. The as-prepared microporous CNTs were further functionalized with an NH₂ group as reported before.^[32,33] The Fourier transform infrared (FTIR) spectra of CNTs and CNTs-NH₂ are shown in Figure S3 (the Supporting Information); the CNT-NH₂ exhibits IR bands in the ranges of 3300–3500 and 1340–1020 nm^{−1} due to stretching vibrations of N–H and C–N, respectively,^[34] suggesting the grafting of NH₂ groups on CNTs.^[35]

Figure 2A–C shows the TEM images of Pt-in-CNTs at different magnifications. We could see that almost all the Pt NPs are nicely dispersed inside the CNTs even if the end of the nanotube was closed. This phenomenon indicates that

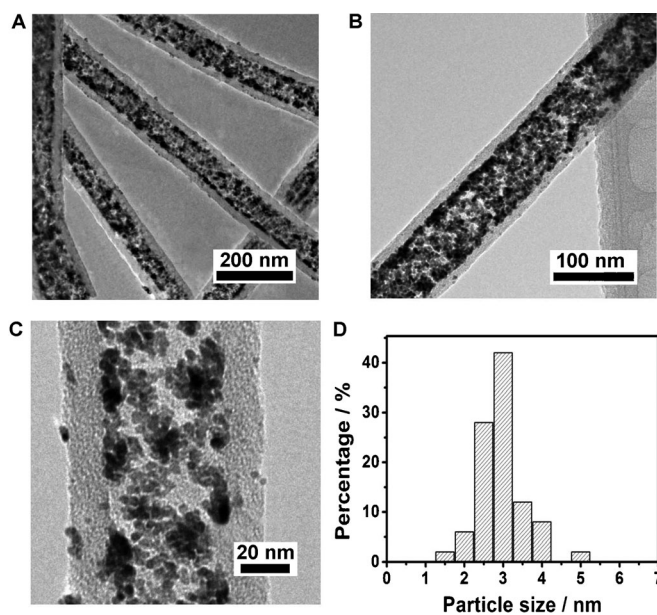


Figure 2. A)–C) TEM images of Pt-in-CNTs at different magnifications. D) Particle-size distribution of Pt NPs in Pt-in-CNTs.

the micropores in CNTs enable the Pt precursors to penetrate into the cavity of CNTs. From the magnified TEM in Figure 2C, the Pt NPs inside CNTs show a diameter distribution in the range from 2 to 5 nm (Figure 2D), which has been suggested as the optimal diameter for catalytic applications.^[35] X-ray photoelectron spectroscopy (XPS) of Pt-in-CNTs is shown in Figure S4 (the Supporting Information). The significant Pt4f, N1s, and C1s signals correspond to the binding energy of Pt, N, and C, respectively, further supporting that the NPs in the CNTs are Pt. ICP-MS was further used to determine the loading amount of Pt NPs inside the CNT. We found that 2 mg CNTs and 7.5 mL of 1.75 mM [PtCl₆]^{2−} resulted in a loading amount of 53.2 % of Pt NPs inside the microporous CNTs.

We can easily control the amount of Pt NPs inside CNTs through simply changing the concentration of Pt precursor. When the concentration is 1.4 mM, Pt NPs were confined in the CNTs channels with a moderate loading amount of 34.8 %, as shown in Figure S5A and B (the Supporting Information). However, as the concentration of H₂[PtCl₆] reaches 2 mM, high-density Pt NPs were distributed on both the internal and external surfaces of CNTs with a Pt loading amount of 69.8 % (Figure S5C and D, the Supporting Information). On the basis of the above results, we deduced that the formation of Pt NPs preferentially occurs inside the channel of CNTs, not outside. When the inside space of CNTs were almost completely occupied by Pt NPs, the Pt NPs would also grow on the surface of CNTs (Figure S5C and D, the Supporting Information). We also found that without any functionalization on CNTs, Pt NPs could only be grown on the surface, not on the inside of CNTs (Figure 3). As shown in the magnified TEM in Figure 3B, the diameter of NPs on Pt-out-CNTs is 2–5 nm, similar to that of Pt-in-CNTs. This is because there are many oxygen-

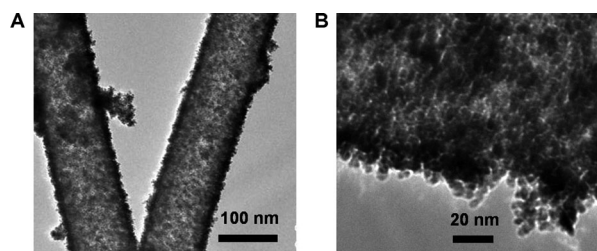


Figure 3. TEM images of Pt-out-CNTs at different magnifications.

containing groups on the surface of CNTs, as revealed by the FTIR data (Figure S3, the Supporting Information). The strong electrostatic repulsions between $-\text{COOH}$ groups and $[\text{PtCl}_6]^{2-}$ impedes the diffusion of $[\text{PtCl}_6]^{2-}$ into the inner cavity of CNTs, and the oxygen-containing groups may accelerate the nucleation of Pt NPs.^[17] In this case, the nucleation of Pt should prefer to happen on the external surface of CNTs.

The interesting structure of Pt-in-CNTs inspired us to investigate their electrocatalytic activities for the MOR. The MOR activity and stability of as-synthesized Pt-in-CNTs and Pt-out-CNTs (Pt-loading amount is 55.3%) were studied by using cyclic voltammetry (CV) and further compared to the benchmarked commercial Pt/C catalyst (20% Pt supported on Vulcan XC-72 carbon, Johnson Matthey). To perform these tests, the catalysts were redispersed in water containing Nafion (0.2 wt%) to reach a concentration of 1 mg mL^{-1} . This dispersion (10 μL) was deposited on the surface of glassy carbon (GC, 3 mm) electrode. The electrochemical surface areas (ECSA) were measured by using hydrogen adsorption/desorption methods in conjunction with CVs of the catalysts in N_2 -saturated $0.5 \text{ M H}_2\text{SO}_4$ solution at a scan rate of 25 mV s^{-1} (Figure 4A). Pt-in-CNTs ($52.5 \text{ m}^2 \text{ g}^{-1}$) show a higher ECSA value than Pt-out-CNTs ($48.7 \text{ m}^2 \text{ g}^{-1}$) and a commercial Pt/C catalyst ($28.3 \text{ m}^2 \text{ g}^{-1}$).

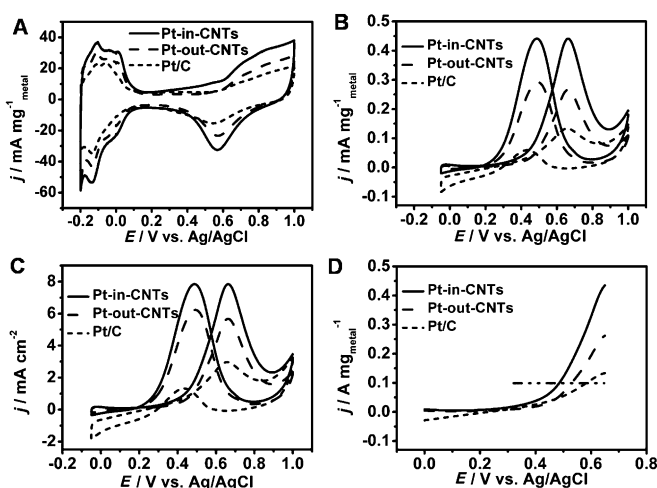


Figure 4. A) CVs of Pt-in-CNTs, Pt-out-CNTs, and commercial Pt/C catalysts in an N_2 -saturated $0.5 \text{ M H}_2\text{SO}_4$ solution at 25 mV s^{-1} ; B, C) CVs and D) linear sweep voltammetry for MOR in $0.5 \text{ M H}_2\text{SO}_4$ solution containing $0.5 \text{ M CH}_3\text{OH}$ at a scan rate of 50 mV s^{-1} .

This result reveals that the Pt-in-CNTs are electrochemically more accessible.

Figure 4B depicts the CVs (mass activity) of Pt-in-CNTs-, Pt-out-CNTs-, and commercial Pt/C catalyst-modified GC electrodes in $0.5 \text{ M H}_2\text{SO}_4$ solution containing 0.5 M methanol at a scan rate of 50 mV s^{-1} . Among the three kinds of catalysts investigated, Pt-in-CNTs shows the highest mass activities than that of the Pt-out-CNTs and the commercial Pt/C catalyst. The mass activity of Pt-in-CNTs is 1.7 and 3.5 times higher than those of Pt-out-CNTs and commercial Pt/C catalyst, respectively. The specific activities of the MOR on Pt-in-CNT, Pt-out-CNT, and a commercial Pt/C were also compared, as shown in Figure 4C. It was found that Pt-in-CNTs have a higher specific activity than those of Pt-out-CNTs and C/Pt, further demonstrating that Pt-in-CNTs have a significantly enhanced electrocatalytic activity. Furthermore, as indicated by the dotted lines in Figure 4D, the corresponding potential of Pt-in-CNTs are lower than that of Pt-out-CNTs, and much lower than that of commercial Pt/C catalyst at a given oxidation current density. In addition, the ratio of the forward oxidation current peak (I_f) to the reverse current peak (I_b), I_f/I_b , is an index of the catalyst tolerance to the poisoning species, $\text{Pt}=\text{C}=\text{O}$.^[36] A higher ratio indicates more effective removal of the poisoning species on the catalyst surface.^[35] Pt-in-CNTs have a higher I_f/I_b value (1.02) than Pt-out-CNTs ($I_f/I_b=0.93$), indicating that Pt-in-CNTs exhibits a better poisoning-tolerance property than Pt-out-CNTs.

A CV cycling technique was applied to study the durability of Pt-in-CNTs, Pt-out-CNTs, and a commercial Pt/C catalyst in $0.5 \text{ M H}_2\text{SO}_4$ solution containing 0.5 M methanol by cycling the potentials between -0.05 and 1 V at the scan rate of 50 mV s^{-1} . As shown in Figure 5A, Pt-in-CNTs display

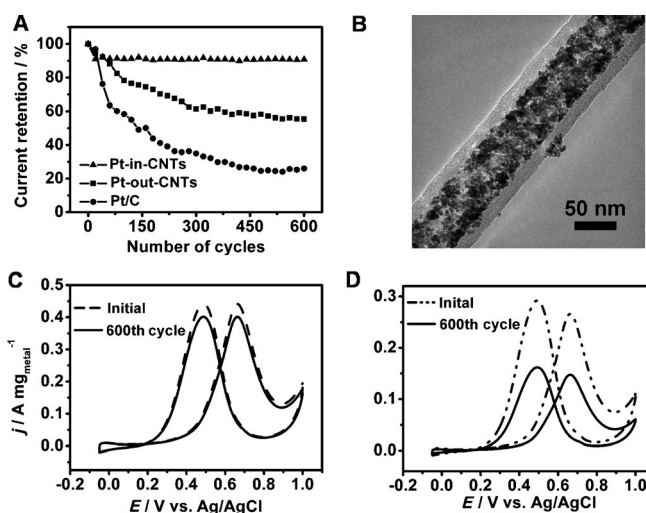


Figure 5. A) Electrocatalytic cycling stability of Pt-in-CNTs, Pt-out-CNTs, and a commercial Pt/C catalyst toward MOR as a function of the number of potential cycles. Current retention values were obtained through normalizing the currents of different cycles to the first one; B) TEM image of Pt-in-CNTs after 600 cycles of MOR; CVs of MOR at C) Pt-in-CNTs and D) Pt-out-CNTs before and after the durability test.

much better durability than those of Pt-out-CNTs, and commercial Pt/C after 600 sequential cycles. Only a loss of 9.1 % was detected in the current density for Pt-in-CNTs after 600 sequential cycles (the corresponding CV changes before and after 600 cycles are shown in Figure 5C). In contrast, there is a big decrease in the current density for Pt-out-CNTs (44.6 % loss, Figure 5D) and a dramatic decrease for the commercial Pt/C catalyst (73.5 % loss) after 600 sequential cycles. We also investigated the morphology changes of Pt-in-CNTs and Pt-out-CNTs before and after the stability test. TEM analysis shows that there is no visible morphology change on Pt-in-CNTs before (Figure 2A–C) and after the stability test (Figure 5B and Figure S6, the Supporting Information). However, in the case of Pt-out-CNTs, Pt NPs tend to congregate and become larger-sized particles during the stability test (Figure S7, the Supporting Information). These results reveal that the confinement of Pt NPs inside the channel of CNTs plays a very important role in greatly enhancing the stability of the MOR.

The results from both the activity and stability studies indicate that the confinement of CNTs has an important role on the MOR catalysis enhancement in both the stability and activity of Pt NPs. Such confinement is not only advantageous for enhancing the electron transfer between CNTs and Pt NPs (enhancing the MOR activity), but also makes it difficult for NPs to move inside CNTs^[16–18,38] and therefore for Ostwald ripening to occur (enhancing MOR stability).^[39,40] The present Pt-in-CNTs exhibited the most stability for MOR catalysts than ever reported before.^[40–43] Furthermore, the adsorption property of tubular nanostructure, especially the capillary force of the CNTs with a microporous wall should allow an increasing concentration of methanol molecules around the active Pt NPs in the nanochannels, which may be also important for promoting the catalytic activity.^[37]

Conclusion

We have developed a controllable strategy for the high-efficiency encapsulation of Pt NPs into the channel of CNTs. More importantly, we have demonstrated that Pt-in-CNTs show remarkable electrochemical performance, with higher electrocatalytic and activity stability (active retention of 91 % after 600 cycles) towards the MOR than Pt-out-CNTs and a commercial Pt/C catalyst. These can be ascribed to the following features: 1) The high conductivity and porous structure of microporous CNTs serve as a highway for electron transfer and easy access for electrolyte ions to the electrode surfaces. This makes it an ideal supporter for a catalyst. 2) The well-ordered channels of CNTs supply a regular molecular- (CH_3OH in our case) and ion-diffusion passway, thus leading to high MOR activity. 3) Good contact between Pt NPs and the internal surface of CNTs produced by the confinement effect of CNTs ensures a tight adhesion between them and avoids Ostwald ripening. Consequently, it can survive corrosive MOR conditions and long-term se-

quential use. On the basis of these intriguing features, the high-performance electrochemical behavior of the Pt-in-CNTs makes it a promising candidate as an anode catalyst for MOR. Furthermore, we expect that the reported filling method can be generally extended to produce various inorganic/organic nanomaterials encapsulated inside the channel of CNTs for catalytic and other applications.

Experimental Section

Synthesis of microporous CNTs: Ag nanowires (NWs) were prepared according to the reported method.^[21] Carbon nanotubes (CNTs) were synthesized by a modified-Stöber method using resorcinol-formaldehyde (RF) resins as precursor and Ag NWs as a template. For a typical synthesis of Ag/RF resins nanocables, Ag NWs (54 mg) were dispersed in a mixture of ethanol (2 mL), ammonia (10 μL), and deionized water (2 mL). Then, resorcinol (10 mg) and formaldehyde (14 μL) were added. After ultrasonic irradiation for 30 min, the resulted reddish-brown suspension was poured into a Teflon-lined stainless-steel autoclave and heated at 100 °C for 24 h. The product was washed and centrifuged several times with ethanol and deionized water, followed by drying at room temperature. The as-prepared Ag/RF resins nanocables were heated under an N_2 atmosphere at 350 °C for 2 h with a heating rate of 1 °C min⁻¹, and subsequently heated at 850 °C for 4 h with a heating rate of 1 °C min⁻¹. After the Ag core was removed with 2 M HNO_3 (50 mL), microporous CNTs were obtained.

Functionalization of CNTs with NH_2 : The as-prepared CNTs (8 mg) were dissolved in ethanol (20 mL), followed by addition of water (2 mL), aqueous ammonia (2 mL), and (3-aminopropyl)triethoxysilane (APTES, 275 μL). After stirring at 60 °C for 8 h, CNTs functionalized with a NH_2 group (CNTs- NH_2) were collected by centrifugation, and then dissolved in water (4 mL).

Synthesis of Pt-in-CNTs and Pt-out-CNTs hybrid nanostructures: For the synthesis of Pt-in-CNTs, the above CNTs- NH_2 solution (1 mL) was mixed in an aqueous solution of $\text{H}_2[\text{PtCl}_6]$ (7.5 mL of 1.75 mM) under ultrasonic irradiation for 30 min at room temperature. After heating to boiling, 0.01 % trisodium citrate (1.5 mL) and ascorbic acid (1 mL of 0.1 M) were added dropwise to the above solution, and were left to react for another 5–10 min. For the synthesis of Pt-out-CNTs, non-modified CNTs (2 mg) were dispersed in an aqueous solution of $\text{H}_2[\text{PtCl}_6]$ (7.5 mL of 2 mM). After heating to boiling, trisodium citrate (1.5 mL of 0.01 %) and ascorbic acid (1 mL of 0.1 M) were rapidly added to the above solution, and allowed to react for another 5–10 min. All these as-synthesized products were centrifuged and washed several times with deionized water and dissolved in water (2 mL).

Characterization: Transmission electron microscopy (TEM) images were collected on a Tecnai G2 20 s-twin with accelerating voltage of 200 kV. Scanning electron microscopy (SEM) images were taken with a field-emission scanning electron microscope (JEOL 7500F) operating at 5 kV. X-ray diffraction (XRD) pattern of the product was recorded on a Rigaku D/MAX-3B using $\text{Cu}_{\text{K}\alpha}$ radiation at a scanning speed of 10 °(2 θ) min⁻¹ in the range of 30–80 °. The thermal decomposition behavior of the samples was monitored by using a Mettler Toledo TGA/DSC 1 analyzer from 30 to 800 °C under air with a heating rate of 10 °C min⁻¹. The nitrogen sorption experiments were carried out at 77 K on a Micromeritics Tristar2200 system with micropore analysis. The Brunauer–Emmett–Teller (BET) specific surface area was calculated using adsorption data at a relative pressure range of $P/P_0 = 0.05$ –0.25. The pore volumes were estimated from the amounts adsorbed at a relative pressure (P/P_0) of 0.99. The micropore-size distribution was calculated with the Horvath–Kawazoe (H–K) method. The X-ray photoelectron spectroscopy (XPS) experiments were performed with an ESCA LAB 250 spectrometer by using a focused monochromatic $\text{Al}_{\text{K}\alpha}$ ($h\nu = 1486.6$ eV) X-ray beam with a diameter of 200 μm . ICP-MS data were collected on an

Aglient 7500a ICP-MS spectrometer. FTIR spectra were obtained on Bruker VerTEX 70 FTIR spectrometer.

Electrocatalytic experiments: Electrochemical experiments were performed on a CHI 660D electrochemical analyzer (CH Instruments, Chenhua Co., Shanghai, P.R. China). A conventional three-electrode cell was used, including a Ag/AgCl (saturated KCl) electrode as reference electrode, a platinum wire as counter electrode, and a grassy carbon (GC) electrode (3 mm in diameter) as a working electrode. For methanol oxidation reaction, the Pt/CNTs hybrids or Pt/C (20% Pt supported on Vulcan XC-72 carbon, Johnson Matthey) catalyst were redispersed in water containing Nafion (0.2 wt%) to reach a concentration of 1 mg mL⁻¹. 10 μ L of this dispersion was deposited on the surface of the pretreated GC electrode and dried before initiating the electrochemical experiments. The methanol oxidation measurements were carried out in a 0.5 M H₂SO₄ solution containing 0.5 M methanol at a scan rate of 50 mV s⁻¹.

Acknowledgements

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